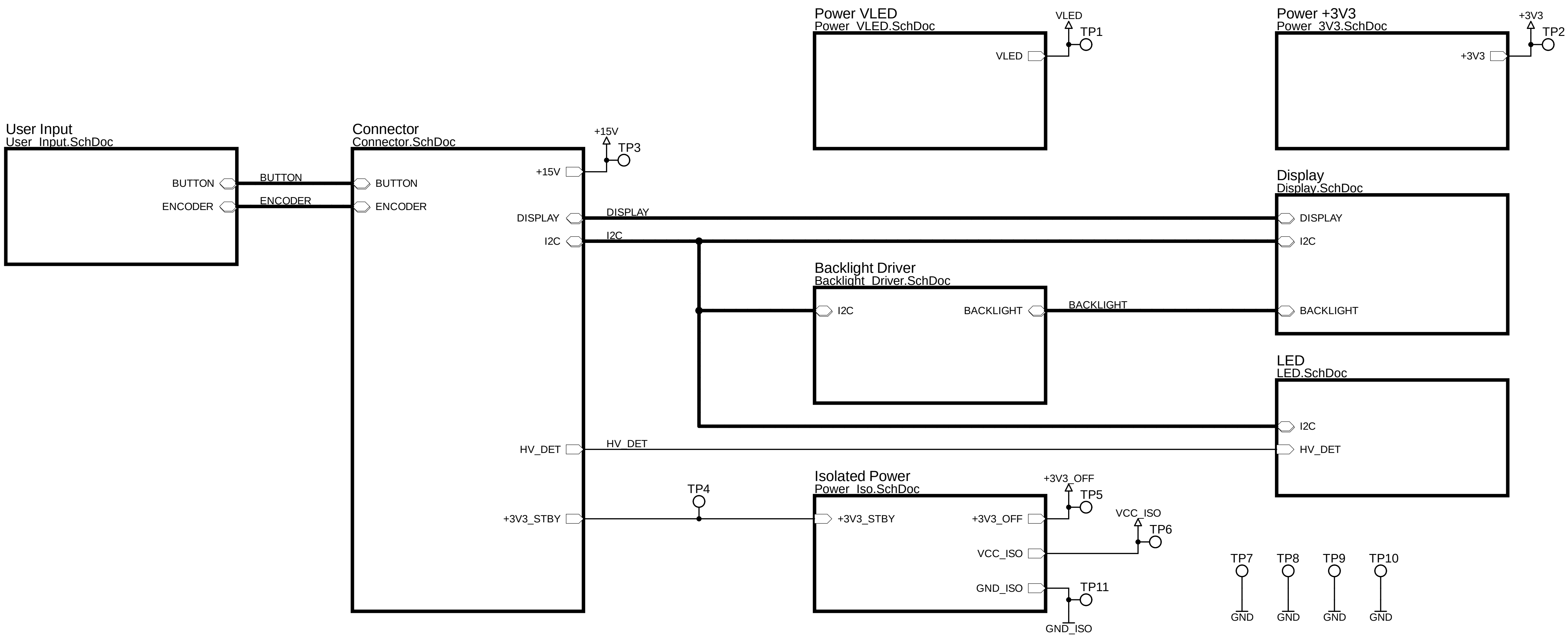
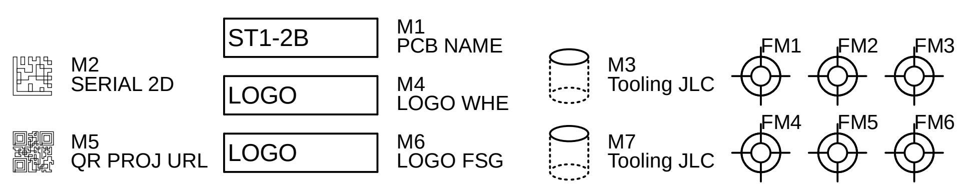
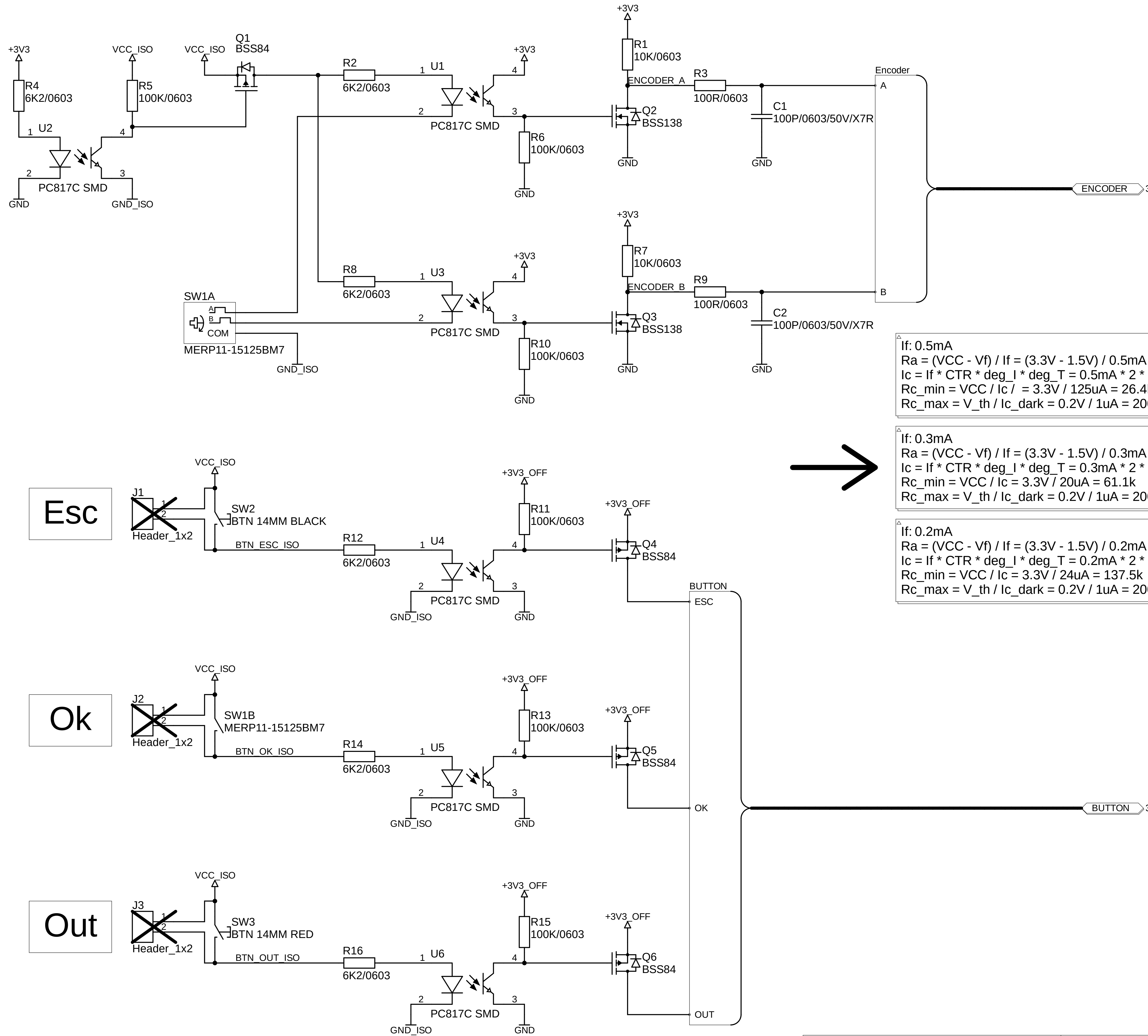




Top		*
ST1-2BB	Scrutester - User Interface	*
Date: 2026-Sep-08	Sheet 1 of 9	Size: A3 *
Schema: ahorat / daniw	Layout: daniw	ST1-2-Top.SchDoc



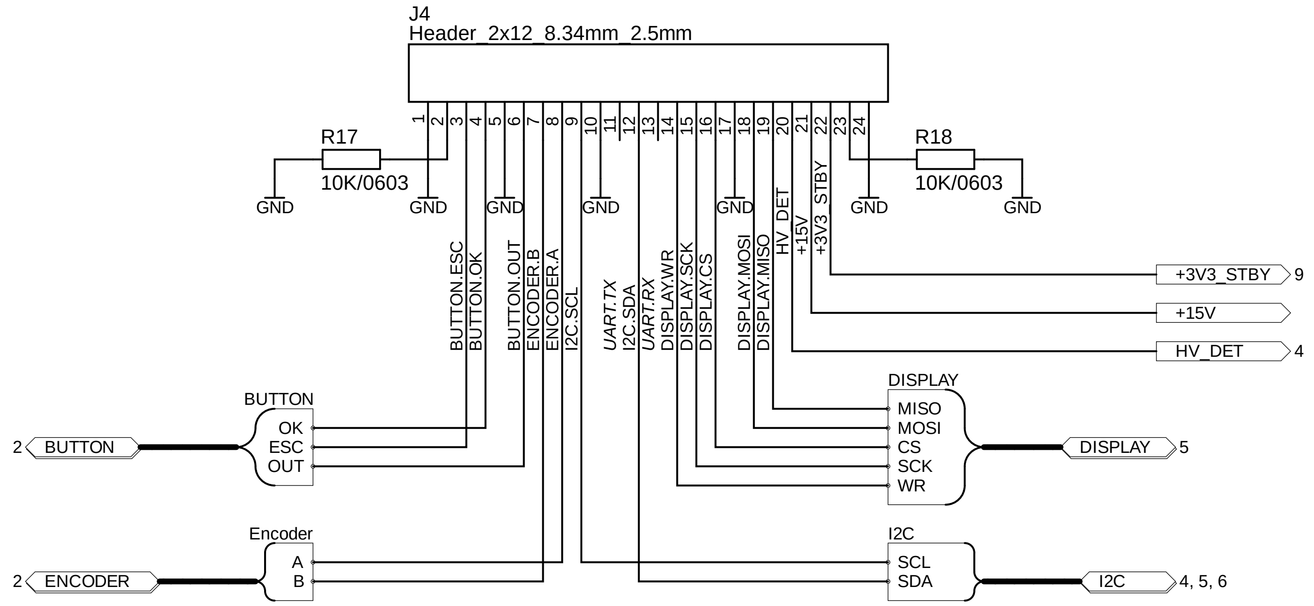
Adding additional optocouplers might be necessary due to isolation requirements for 600V CAT III (11mm)

If: 0.5mA
Ra = (VCC - Vf) / If = (3.3V - 1.5V) / 0.5mA = 3.6k
Ic = If * CTR * deg_I * deg_T = 0.5mA * 2 * 0.25 * 0.5 = 125uA
Rc_min = VCC / Ic / = 3.3V / 125uA = 26.4k
Rc_max = V_th / Ic_dark = 0.2V / 1uA = 200k

If: 0.3mA
Ra = (VCC - Vf) / If = (3.3V - 1.5V) / 0.3mA = 6k
Ic = If * CTR * deg_I * deg_T = 0.3mA * 2 * 0.18 * 0.5 = 54uA
Rc_min = VCC / Ic = 3.3V / 20uA = 61.1k
Rc_max = V_th / Ic_dark = 0.2V / 1uA = 200k

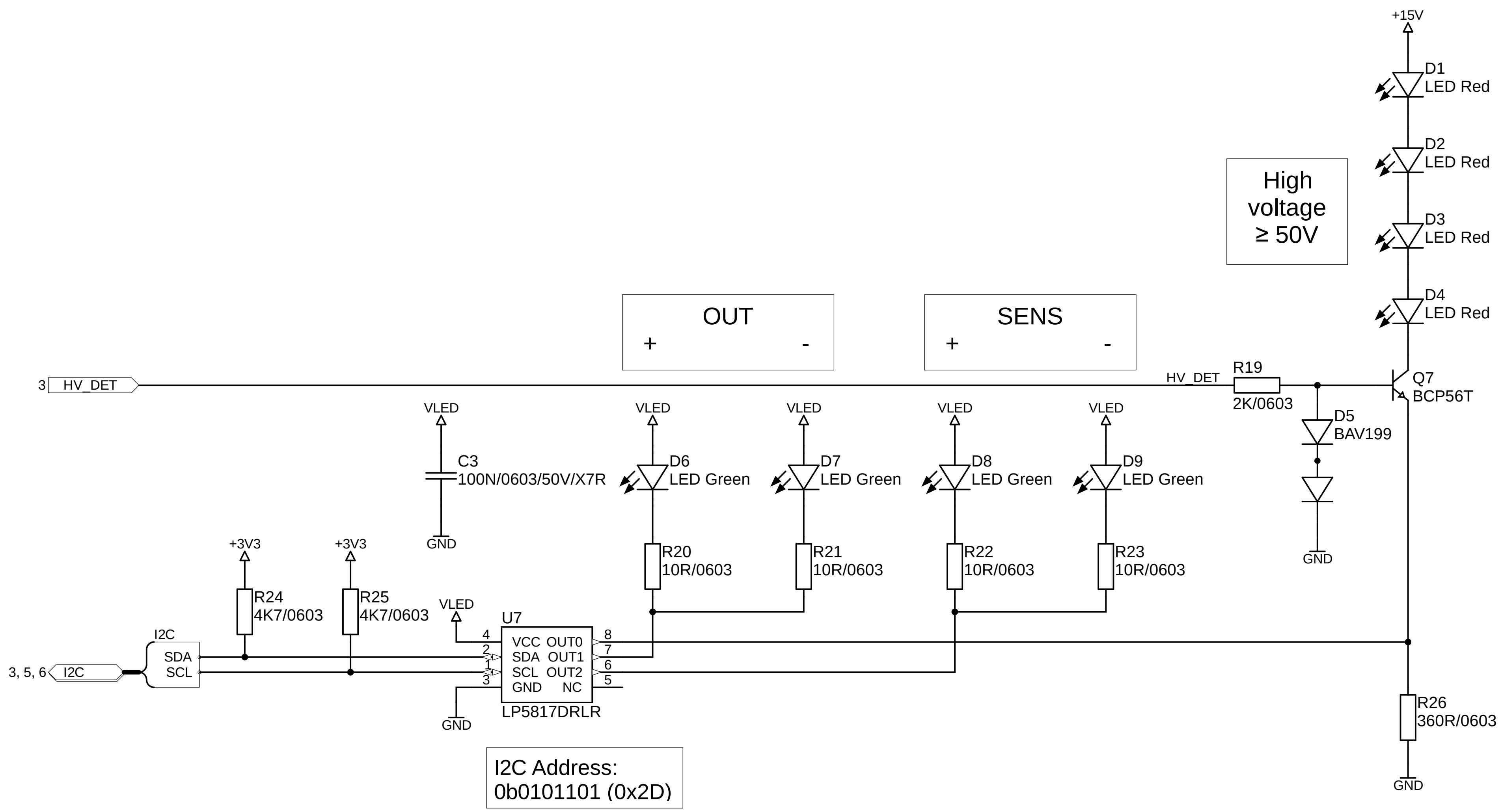
If: 0.2mA
Ra = (VCC - Vf) / If = (3.3V - 1.5V) / 0.2mA = 9k
Ic = If * CTR * deg_I * deg_T = 0.2mA * 2 * 0.12 * 0.5 = 24uA
Rc_min = VCC / Ic = 3.3V / 24uA = 137.5k
Rc_max = V_th / Ic_dark = 0.2V / 1uA = 200k

User Input		*
ST1-2BB	Scrutester - User Interface	*
Date: 2026-Sep-08	Sheet 2 of 9	Size: A3 *
Schema: ahorat / daniw	Layout: daniw	User_Input.SchDoc

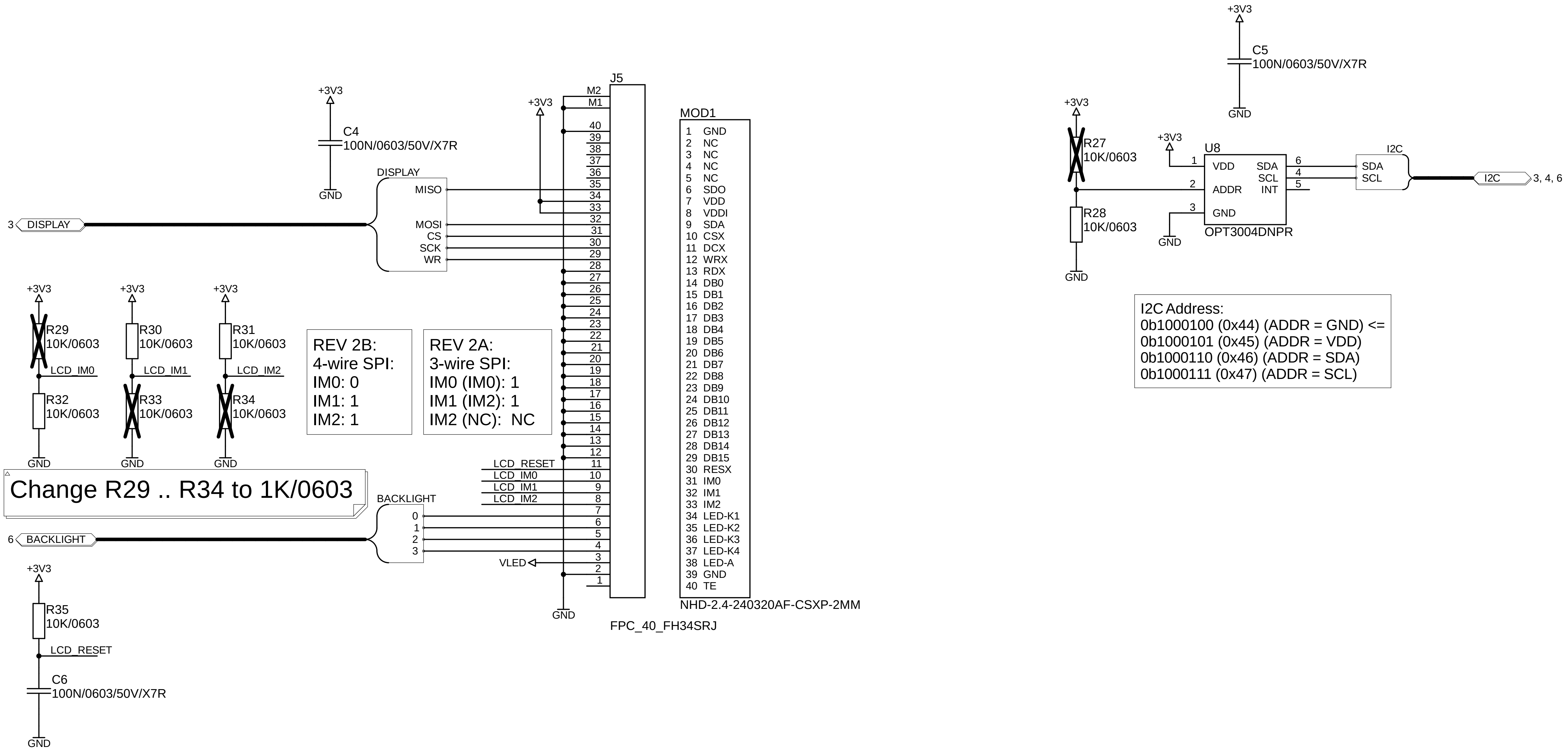


Connector

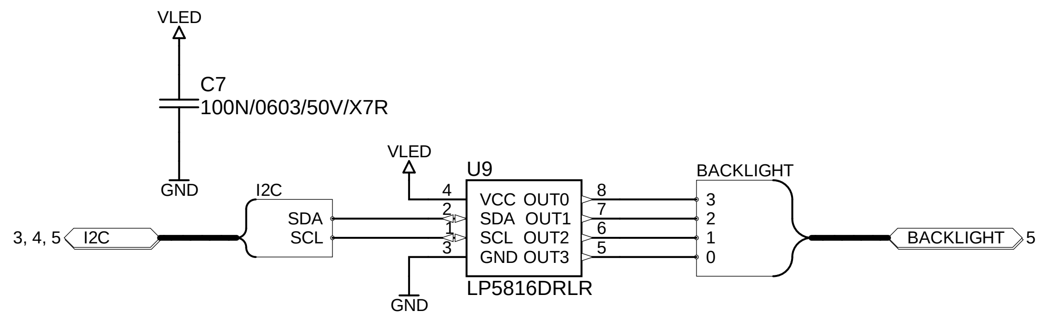
ST1-2BB	Scrutester - User Interface	*	
Date: 2026-Sep-08	Sheet 3 of 9	Size: A3	
Schema: ahorat / daniw	Layout: daniw	Connector.SchDoc	



LED			
ST1-2BB	Scrutester - User Interface		*
Date: 2026-Sep-08	Sheet 4 of 9	Size: A3	*
Schema: ahorat / daniw	Layout: daniw	LED.SchDoc	*



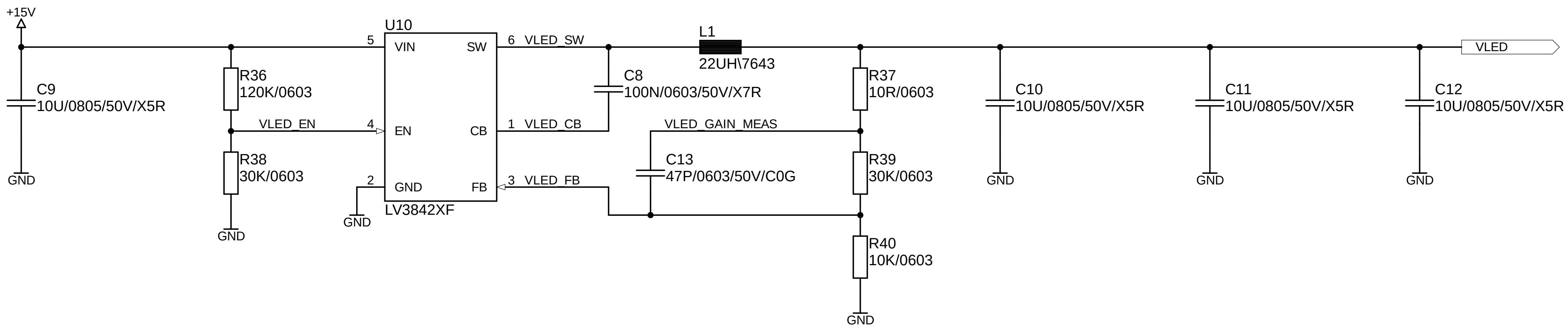
Display		*	
ST1-2BB	Scrutester - User Interface	*	
Date: 2026-Sep-08	Sheet 5 of 9	Size: A3	*
Schema: ahorat / daniw	Layout: daniw	Display.SchDoc	



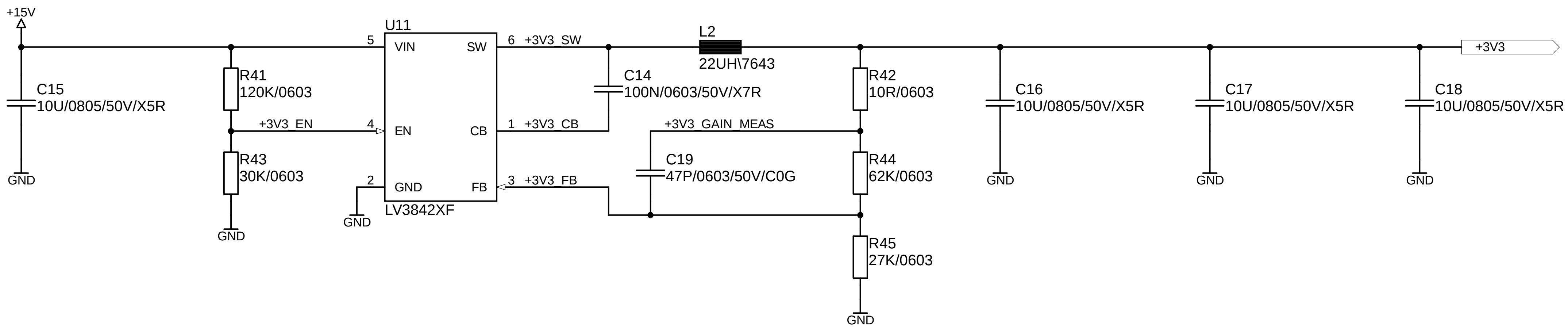
I2C Address:
0b0101100 (0x2C)

Maximum channel
current: 40mA

Backlight Driver		*
ST1-2BB	Scrutester - User Interface	*
Date: 2026-Sep-08	Sheet 6 of 9	Size: A3
Schema: ahorat / daniw	Layout: daniw	Backlight_Driver.SchDoc

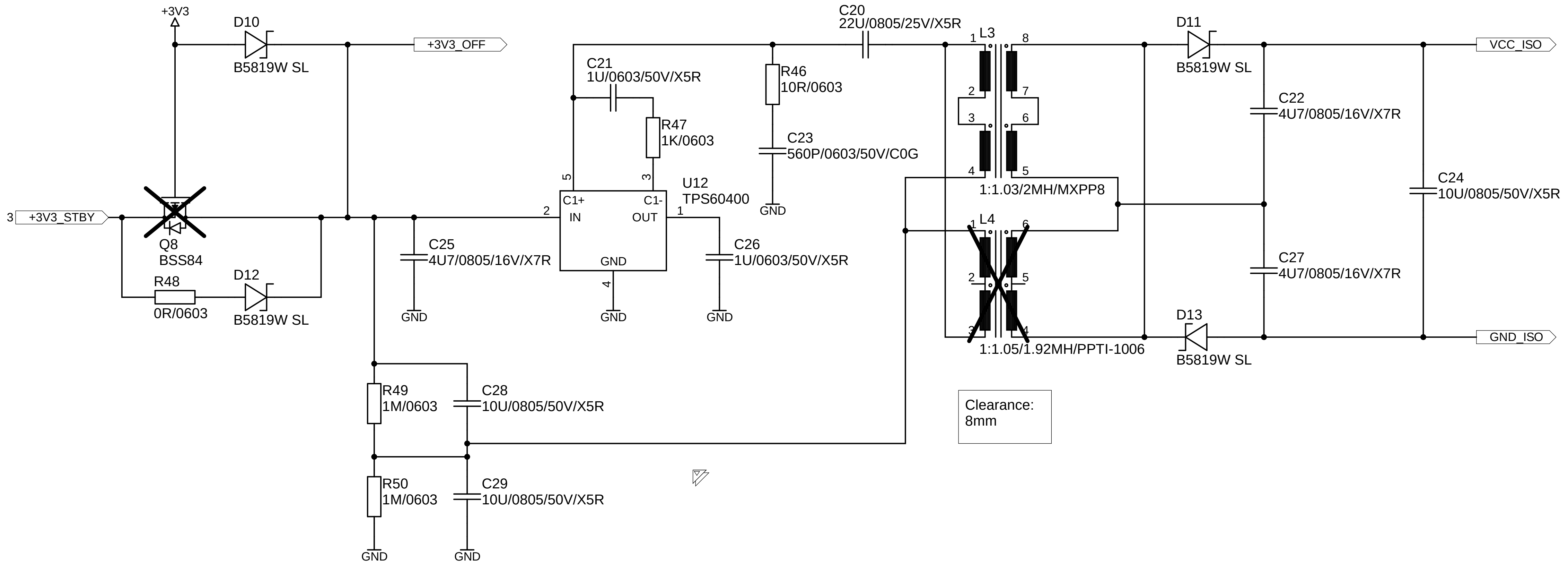


Power VLED			*
ST1-2BB	Scrutester - User Interface		*
Date: 2026-Sep-08	Sheet 7 of 9	Size: A3	*
Schema: ahorat / daniw	Layout: daniw	Power_VLED.SchDoc	



Power +3V3			*
ST1-2BB	Scrutester - User Interface		*
Date: 2026-Sep-08	Sheet 8 of 9	Size: A3	*
Schema: ahorat / daniw	Layout: daniw	Power_3V3.SchDoc	

Adding additional transformers might be necessary due to isolation requirements for 600V CAT III (11mm)



Clearance:
8mm

Power isolated		*
ST1-2BB	Scrutester - User Interface	*
Date: 2026-Sep-08	Sheet 9 of 9	Size: A3
Schema: ahorat / daniw	Layout: daniw	Power_Iso.SchDoc